END-TO-END FLOOD FORECASTING & ANTICIPATORY ACTION PIPELINE

Operational Multi-Head GBDT Engine (Task A) ⟶ Area-Preserving ~232 m Downscaler (Task B) ⟶ Multi-Asset Exposure Action

1. DATA INGESTION & EMBARGO

STRICT FRIDAY 00:00 UTC OPERATIONAL EMBARGO
Zero lookahead leakage; mirrors Red Cross anticipatory cycle

Upstream Hydrometeorology (ERA5-Land)

- Catchment Runoff (Q_upstream): 1, 7, 14, 30-day lags
- Net Water Surplus: P - ET_0 cumulative balance
- Spatial Catchment: ~300 km Congo-Nile divide transit

Remote Water Levels & River Gauges

- River Lol Stream Gauges: gauge_w1, w7, w14 & 7d change
- Lake Albert & Sudd Stage Altimetry Signals
- Hydraulic memory provides 2–4 week predictive lead

Historical Satellite Baseline (MODIS)

- 26-Year Historical Record (2000–2025 Terra/Aqua)
- ~232 m Native Resolution 8-day composite masks
- Establishes baseline flood frequency & dry contours

Lagged Vector →

2. MULTI-HEAD GBDT ENGINE (LIGHTGBM)

Feature Ingestion & Autoregression

Hydrometeorological Lags + Harmonic Seasonality (sin/cos)
+ Autoregressive Flood Extent (y_lag1, y_lag2)

Head 1: Flood Detection Classifier [P(flood > 0)]

Binary LogLoss | Predicts county inundation probability with 84.7% accuracy

Head 2: County Flood Extent Regressor [q10, q50, q90 km²]

Pinball Quantile Loss | Calibrated 10th, 50th, 90th percentile area bounds

Head 3: Flood Persistence / Duration [h1, h2, h3, h4 weeks]

Multi-Horizon Survival | Tracks flood persistence across 1 to 4 week lead horizons

Optuna Bayesian Hyperparameter Optimization

50+ trials over lr, depth, & L1/L2 regularization identify physical saturation
at the irreducible observational noise floor (loss ~0.366).

County q50 (km²) →

3. DOWNSCALING & EXPOSURE

Area-Preserving Pixel Disaggregation (Task B)

- Step 1: Ingest tabular county median extent q50 (km²)
- Step 2: Conditional pixel climatology P(pixel | month, county)
- Step 3: Discrete step merge: k = round(v / step - 0.5)
- Area Conservation: Σ sev_p × A_p ≡ q50 (Exact Δ = 0.000 km²)
- Native ~232 m satellite grid inundation probability map

[CROPLAND EXPOSURE] Stationary Food Security

- ASAP Crop Mask Overlay (Sorghum / Groundnuts)
- Critical Ripening / Harvest Window (Sep–Nov)
- Anticipatory Action: 2–3 week warning triggers
  emergency harvesting & protective drainage clearing

[CATTLE EXPOSURE] Mobile Pastoral Livestock

- GeoNode Gridded Livestock Density Overlay
- Floodplain Grazing Depressions & Transhumance Paths
- Anticipatory Action: 7–14 day warning triggers
  coordinated dry corridor migration & livestock vaccines